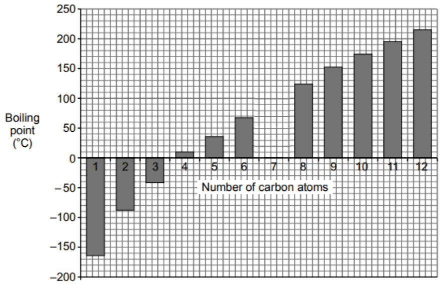
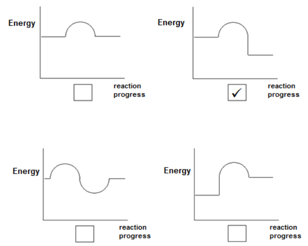
## Common questions

| Question | | | | Marking details | Marks available | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | AO1 | AO2 | AO3 | Total | Maths | Prac |
| **6/1** | *(a)* | | | the larger the molecules / (as the size/fraction) gets bigger<br>• the smokier the flame (1)<br>• the more difficult it becomes to burn (1)<br><br>accept the converse argument<br><br>correct identification of both properties, without reference to increasing size (1) | | | 2 | 2 | | |
| | *(b)* | (i) | | <br>all 5 bars correctly plotted with ±½ small square tolerance (2)<br>3 or 4 correct plots (1)<br><br>accept charts where bars are touching – correct height of each bar to be credited | | 2 | | 2 | 2 | |

| Question | | | | Marking details | Marks available | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | AO1 | AO2 | AO3 | Total | Maths | Prac |
| | | (ii) | | appropriate **straight** trend line drawn with ruler, spans across at least 6 bars (1)<br><br>correct boiling point taken from the trend line (1)<br><br>award no credit for a curved trend line (but allow ECF for a correct value read from an incorrect trend line)<br><br>if no trend line drawn, award (1) for a value in the range 85-105 | | | 2 | 2 | 2 | |
| | *(c)* | (i) | | sand / foam / $CO_2$ / fire blanket<br>because it removes oxygen<br>**both** method and explanation needed | | 1 | | 1 | | |
| | | (ii) | | $5CO_2$ + $6H_2O$<br><br>correct products (1)<br>correctly balanced (1) – only if correct products given | | 2 | | 2 | 1 | |
| | | (iii) | | does not produce carbon dioxide / sulfur dioxide / **<u>only</u>** produces water (1)<br><br>does not contribute to global warming / climate change / acid rain (1) | 1 | 1 | | 2 | | |

| Question | | | | Marking details | Marks available | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | AO1 | AO2 | AO3 | Total | Maths | Prac |
| | *(d)* | (i) | | temperature (of water) **before and after** burning (1)<br>mass of fuel and burner **before and after** burning (1)<br><br>initial ≡ before ≡ at start<br>final ≡ after ≡ at end<br><br>temperature rise and change in mass – neutral answers<br><br>award (1) only for reference to measuring both temperature throughout and mass throughout<br><br>award (1) for answers that refer to measuring **both** the temperature and mass **either** before **or** after burning only | | | 2 | 2 | | 2 |
| | | (ii) | | distance between burner and flame / material or thickness or size of beaker / same size wick / same beaker<br><br>reference to mass of fuel and mass of water – neutral | 1 | | | 1 | | 1 |
| | | (iii) | |  | | 1 | | 1 | 1 | |
| | | | | **Question 6/1 total** | **2** | **7** | **6** | **15** | **6** | **3** |